

Alexander Parks

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EDUCATION

The University of Texas at Austin, Austin, TX

August 2025 - May 2027 (expected)

Master of Science, Computer Science

- GPA: 4.0
- Relevant coursework: Deep Learning, Reinforcement Learning, Case Studies in Machine Learning, Linear Algebra, NLP

The University of Texas at Austin, Austin, TX

August 2021 - May 2025

Bachelor of Science, Computer Science; Bachelor of Science, Mathematics

- GPA: 3.74
- Relevant coursework: Machine Learning I/II, Neural Networks, Trustworthy ML, Probability, Applied/Mathematical Statistics, Predictive Analytics, Linear Algebra, Data Management, NLP, Stochastic Processes, Algorithms and Complexity, Data Structures, Software Engineering, Operating Systems, Computer Architecture, Real Analysis, Differential Equations
- Honors: Eva Woods Endowed Presidential Scholar, Bitar Family Endowed Scholar in Computer Science, University Honors

SKILLS

Data Analysis/Data Science: Python, R, RStudio, pandas, scikit-learn, TensorFlow, PyTorch, NumPy, Matplotlib, SQL, PostgreSQL, MySQL Workbench, Jupyter, Microsoft Excel, Hugging Face Transformers

Software Engineering: Java, C, TypeScript, HTML, CSS, React, Flask, Node.js, Postman, Jira, SQLAlchemy, Playwright, pytest

Cloud: Amazon Web Services (AWS), Google Cloud Platform (GCP), Cloudflare

Miscellaneous: Git, GitHub, GitLab, Visual Studio Code, Microsoft Word, Microsoft PowerPoint

EXPERIENCE

Regions Bank, Atlanta, GA

May 2024 - August 2024

Software Engineer Intern

- Developed 5+ new portal components using TypeScript, C, and AWS CLI in an Agile environment for RelationshipIQ, a cloud-native onboarding platform for prospective Regions Bank business clients
- Authored 100+ unit and coverage tests to validate new component behavior and platform integration
- Conducted research on the machine learning and artificial intelligence systems of 10+ peer banks
- Achieved first place in the 2024 Regions Bank AWS GameDay Hackathon

PROJECTS

FBS College Football Score Prediction

- Engineered an end-to-end ML pipeline using CollegeFootballData API data from 9,000+ FBS games, incorporating scoring, efficiency, EPA, success rate, Elo, recruiting talent, and game-context features
- Trained and tuned six models: Ridge, Elastic Net, Random Forest, Gradient Boosting, SVR, and a PyTorch neural network using expanding-season cross-validation and an out-of-fold stacking ensemble
- Evaluated final models on an untouched 808-game 2025 holdout, achieving 72.9% winner accuracy and MAEs of 12.55 for scoring margin and 12.65 for total score

Evaluating Political Bias in Popular Generative LLMs

- Constructed a dataset of 750+ political sentence prefixes by scraping articles from multiple news organizations and generating paired counterfactual prompts with matched political terminology
- Generated sentence completions across several LLMs using Hugging Face Transformers and the OpenAI API to compare behavior across multiple generations of language models
- Built an NLP evaluation pipeline using transformer-based sentiment analysis to quantify and compare sentiment differences between model outputs from paired political prompts

RefLab (password available upon request)

- Built an educational platform for football referees using React and TypeScript with a Python ETL pipeline, creating a searchable 588-node rule corpus from the NCAA Rules Book with hierarchical metadata and linked citations
- Engineered a RAG assistant using custom BM25 retrieval, query expansion, exact-ID matching, and rule graph traversal
- Designed citation-grounding safeguards and a 48-question RAG evaluation suite, achieving 100% citation recall at the top-eight cutoff and 98.68% average citation recall with the final retriever

LEADERSHIP AND COMMUNITY INVOLVEMENT

Computer Science Transfers' Society (CSTS)

August 2021 - May 2025

President (2024 - 2025), Treasurer (2023 - 2024), Member (2021 - 2025)

- Planned and budgeted for transfer application and professional workshops, officer meetings, and socials
- Authored and distributed an annual sponsorship prospectus for current and potential corporate partners